int3ract: Johnson–Neyman Technique and Its Three-Way Extension for Frequentist and Bayesian Models in R

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Interaction effects are ubiquitous in applied statistical modelling, yet their meaningful interpretation remains challenging. The classic Johnson–Neyman (JN) technique (Johnson and Neyman 1936) addresses this challenge for two-way interactions by identifying the regions of a moderator's range over which a focal effect is and is not statistically significant. The **int3ract** package for R implements the JN technique and its three-way extension (the Johnson–Neyman–Krause, or JNK, technique) for both frequentist and Bayesian models. The core function `JNK_int()` can be applied to multiplicative interactions from (virtually) any model family. Dedicated wrappers (`JNK_lm()` and `JNK_siena()`) streamline the workflow for models fitted via `lm()`/`glm()` and **RSiena**'s `siena()`, respectively. For Bayesian Stochastic Actor-Oriented Models (SAOMs) estimated with **multiSiena**, the function `JNSienaBayes()` produces conditional posterior distributions. For two-way interactions, classic shaded confidence-band plots are created that visually demarcate significant and non-significant regions along the moderator range; three-way interactions yield colour-gradient heatmaps with optional crosshatch overlays for non-significant regions. The package is designed to encourage richer, region-specific reporting of interaction effects in place of the conventional single-slope spotlight approach.

1 Introduction

Multiplicative interaction terms allow researchers to examine whether the relationship between a predictor X and an outcome Y depends on the value of one or more additional variables—the so-called moderators. Interpreting such interactions by inspecting the sign and significance of the interaction coefficient alone is, however, insufficient and potentially misleading (Bauer and Curran 2005; Preacher, Curran, and Bauer 2006). The key insight behind the Johnson–Neyman (JN) technique is that instead of evaluating the focal effect at a small number of

selected moderator values (often referred to as "spotlight" or "simple-slope"), one can derive the *entire* region of the moderator's distribution over which the focal effect is statistically distinguishable from zero (Johnson and Neyman 1936; Potthoff 1964).

Originally proposed in the context of the analysis of covariance (Johnson and Neyman 1936), the JN technique was later adapted for regression (Potthoff 1964) and updated by Bauer and Curran (2005), whose formulation is closest to what **int3ract** implements. Several R packages already address the two-way JN case, for instance, **interactions** (Long 2019) and **jtools** (Long 2020). The contribution of **int3ract** is threefold.

1. **Three-way interactions.** The package provides a complete implementation of the JNK technique for three-way multiplicative interactions, producing a heatmap over the two-dimensional moderator grid rather than a single-axis ribbon plot.
2. **Model-agnostic frequentist engine.** The internal function `JNK_int()` requires only a coefficient vector and the relevant sub-matrix of the parameter covariance matrix. It therefore applies to any frequentist model that produces these quantities—ordinary least squares, generalised linear models, mixed-effects models, survival models, or Stochastic Actor-Oriented Models—without modification.
3. **Bayesian SAOMs.** The function `JNSienaBayes()` adapts the technique to the posterior distribution rather than the sampling distribution, producing density plots (two-way) and posterior-mean heatmaps with Bayesian p-value overlays (three-way) for models estimated with **multiSiena**.

The remainder of the paper is organised as follows. Section 2 reviews the statistical background of the JN technique and its three-way extension. Section 3 describes the package architecture. Section 4 and Section 5 illustrate the frequentist and Bayesian workflows with worked examples. Section 6 discusses graphical customisation. Section 7 concludes.

2 Statistical Background

2.1 The Two-Way Johnson–Neyman Technique

Consider the linear model

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 M_i + \beta_3 X_i M_i + \varepsilon_i, \tag{1}$$

where X is the focal predictor, M is the moderator, and β_3 is the interaction coefficient. The simple slope of Y with respect to X at a given value m of the moderator is

$$\hat{\theta}_X(m) = \hat{\beta}_1 + \hat{\beta}_3\, m. \tag{2}$$

Its sampling variance follows from the delta method:

$$\widehat{Var}[\hat{\theta}_X(m)] = \widehat{Var}(\hat{\beta}_1) + 2m\,\widehat{Cov}(\hat{\beta}_1, \hat{\beta}_3) + m^2\,\widehat{Var}(\hat{\beta}_3). \tag{3}$$

The two-way JN regions are the set of moderator values m for which the two-sided z-test

$$z(m) = \frac{\hat{\theta}_X(m)}{\sqrt{\widehat{Var}[\hat{\theta}_X(m)]}}$$

satisfies $|z(m)| > z_{\alpha/2}$, where $z_{\alpha/2}$ is the $(\alpha/2)$-upper quantile of the standard normal distribution.

The symmetry of Equation 1 implies that the roles of X and M are interchangeable: the simple slope of Y on M moderated by X is $\hat{\beta}_2 + \hat{\beta}_3\,x$, with an analogous variance formula. **int3ract** therefore always produces *two* JN plots per two-way interaction and three for three-way interactions – one for each variable acting as the focal predictor.

2.2 The Three-Way Extension

When a third variable W enters as an additional moderator, the full three-way interaction model is

$$\begin{aligned} Y_i = \beta_0 &+ \beta_1 X_i + \beta_2 M_i + \beta_3 W_i + \beta_4 X_i M_i + \beta_5 X_i W_i + \beta_6 M_i W_i \\ &+ \beta_7 X_i M_i W_i + \varepsilon_i. \end{aligned} \tag{4}$$

The conditional effect of X given M and W is

$$\theta_X(m, w) = \beta_1 + \beta_4 m + \beta_5 w + \beta_7 mw, \tag{5}$$

and the corresponding standard error is obtained via the multivariate delta method:

$$\begin{aligned} \widehat{Var}[\hat{\theta}_X(m, w)] = \widehat{Var}_{\beta_1} &+ m^2\widehat{Var}_{\beta_4} + w^2\widehat{Var}_{\beta_5} + (mw)^2\widehat{Var}_{\beta_7} \\ &+ 2m\widehat{Cov}(\beta_1, \beta_4) + 2w\widehat{Cov}(\beta_1, \beta_5) + 2mw\widehat{Cov}(\beta_1, \beta_7) \\ &+ 2mw\widehat{Cov}(\beta_4, \beta_5) + 2m^2w\widehat{Cov}(\beta_4, \beta_7) \\ &+ 2mw^2\widehat{Cov}(\beta_5, \beta_7). \end{aligned} \tag{6}$$

Because the significance region is now two-dimensional, it cannot be depicted as a simple interval on a line. **int3ract** instead renders a heatmap of conditional point estimates $\hat{\theta}_X(m, w)$ using a diverging colour scale, and superimposes a crosshatch pattern over cells where the null

hypothesis $\theta_X = 0$ should be rejected given the chosen α-threshold (default: `threshold = 0.05`).

3 Package Architecture

The package is organised into four layers, summarised in Table 1.

Table 1: Function hierarchy of **int3ract**. User-facing wrappers extract the relevant coefficients and covariance sub-matrix from the model object and pass them to the appropriate engine.

Entry point	Model class	Core engine
`JNK_lm()`	`lm`, `glm`	`JNK_int()`
`JNK_siena()`	`sienaFit` (frequentist)	`JNK_int()`
`JNSienaBayes()`	`multiSiena` (Bayesian)	`jnb_support2()`, `jnb_support3()`
`JNK_int()`	*any* (generic)	`parFun()`, `seFun()`

Utility layer (`utils.R`).
`parFun()` and `seFun()` implement Equation 5 and Equation 6.

Core frequentist engine (`JNK_int()`).
The function `JNK_int()` accepts `name` (character vector of variable names), `coefs` (named numeric vector of model coefficients), `covar` (the corresponding covariance sub-matrix), and `vals` (a list of moderator value ranges). When a 2-way interaction is analysed, it returns two ribbon plots and two data frames; when a 3-way interaction, it returns three heatmaps together with the underlying matrices of point estimates, standard errors, p-values, and significance indicators.

Crucially, `JNK_int()` is model-agnostic. Any researcher who can extract a coefficient vector and covariance matrix from their model (whether fitted by `lm()`, `glmer()`, `coxph()`, or any other estimator) can call `JNK_int()` directly. The function can also be applied to Bayesian estimations, if one would reduce the posteriors to their mean and covariance structure.

Frequentist wrappers.
`JNK_lm()` extracts `coefficients` and `vcov()` from an `lm` or `glm` object and constructs the interaction-term names following R's standard `:` convention before delegating to `JNK_int()`.

`JNK_siena()` extracts `$theta` and `$covtheta` from a `sienaFit` object. Because **RSiena** effects are harder to index by name[1], the user supplies the indices for all main effects and interaction terms.

[1] While effects are identifyable by name, in complex models the same effect can appear multiple times for different dependent variables and/or as different effect types, making it harder on the user to provide

Bayesian engine (`JNSienaBayes()`, `jnb_support2()`, `jnb_support3()`).
For Bayesian SAOMs there is no single-point estimate or analytic covariance matrix. Instead, conditional posteriors will be used. `JNSienaBayes()` handles burn-in removal and thinning, detects whether each parameter was estimated as fixed (η) or random (μ), and routes to `jnb_support2()` for 2-way interactions or `jnb_support3()` for 3-way interactions.

For two-way interactions, `jnb_support2()` computes the conditional posterior draws

$$\tilde{\theta}_X^{(s)}(m) = \theta_X^{(s)} + \theta_{XM}^{(s)} \cdot m$$

at each sampled iteration s and moderator value m, yielding a full conditional posterior distribution. These are visualised as overlapping density curves coloured by moderator value, and summarised by the posterior mean, posterior standard deviation, and Bayesian p-value $\Pr(\tilde{\theta}_X > 0 \mid \text{data}, \text{model}, \text{prior})$.

For three-way interactions, `jnb_support3()` extends this to the grid (m, w), producing heatmaps of both the posterior mean and the Bayesian p-value, each overlaid with a crosshatch on cells outside the user-specified "significance" thresholds.

4 Frequentist Usage

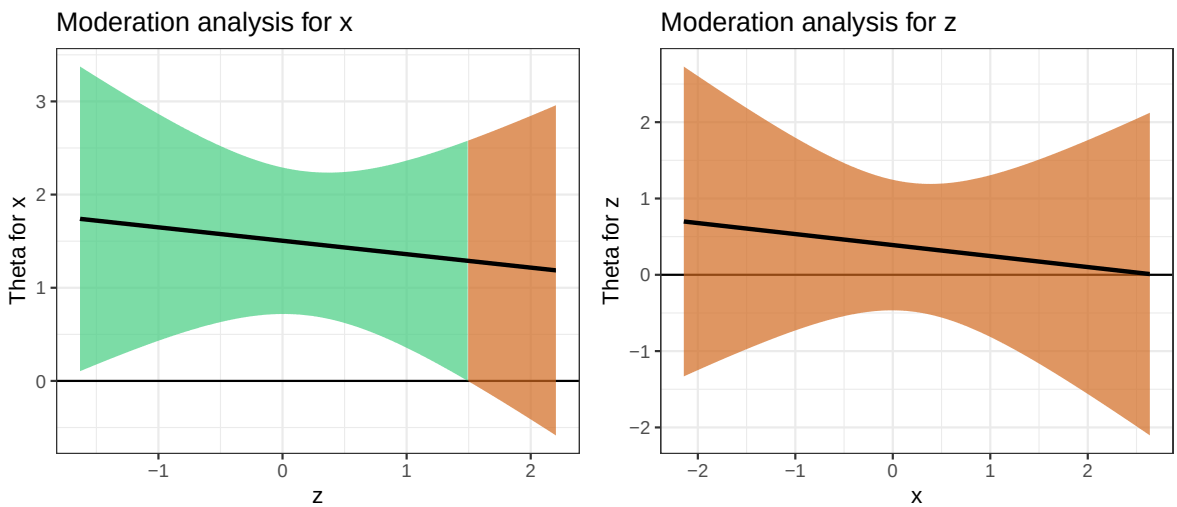
4.1 Two-Way Interaction with `lm`

The simplest entry point is `JNK_lm()` applied to an ordinary least-squares model.

```
library(int3ract)

set.seed(1402)
dat <- data.frame(x = rnorm(100),
                  z = rnorm(100),
                  w = rnorm(100))

dat$y = dat$x + 0.5 * dat$z + -.5 * dat$w + 0.5 * dat$x *
  dat$z * dat$w + rnorm(100,sd = 4)

mod2 <- lm(y ~ x * z, data = dat)

jn2 <- JNK_lm(mod2,
              theta_1 = "x",
```

accurate input to the `JNK_siena()` compared to using the already provided numbering system in `siena()` outputs.

```
                theta_2 = "z",
                alpha   = 0.05)
```



(a) Conditional effect of `x` as a function of the moderator `z`.

(b) Conditional effect of `z` as a function of the moderator `x`.

Figure 1: Johnson–Neyman plots for the two-way interaction between `x` and `z` ($\alpha = 0.05$). The ribbon shows pointwise 95% confidence bands for the conditional effect; green shading indicates regions where the effect is statistically significant, chocolate where it is not.

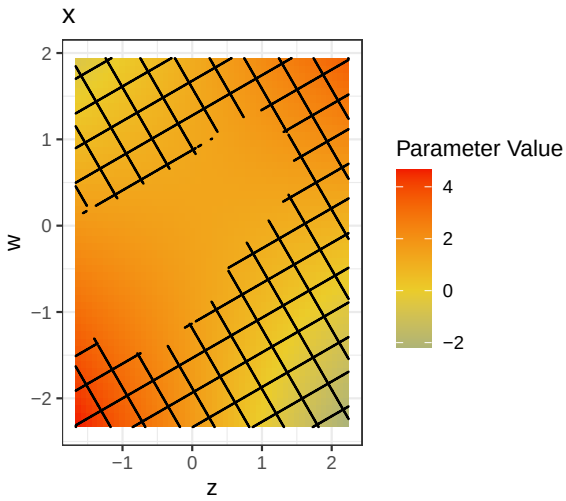
The returned list `jn2` contains `$param_table`, a list of two data frames (one per each variable involved in the interactions) with columns for the moderator values (`mod_vals`), conditional parameter, its standard error, and p-value (`theta_vals`, `theta_se`, and `theta_p`), and a logical whether it is significant given the chosen `threshold`; and `$plots`, a list of two **ggplot2** objects.

In each plot the ribbon is filled with `sig_color` (default `'seagreen3'`) where $|z| > z_{\alpha/2}$ and with `non_sig_color` (default `'chocolate'`) elsewhere.

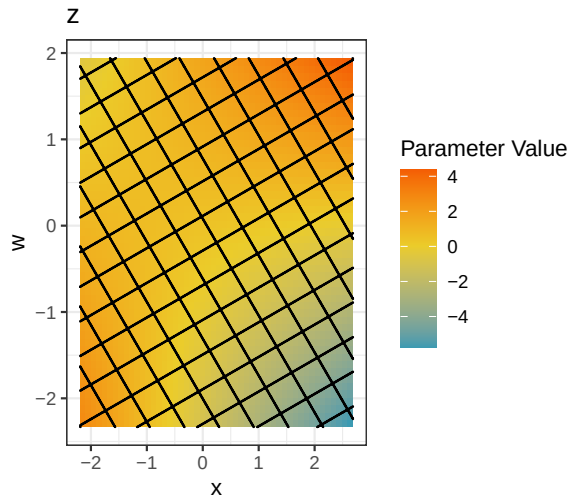
4.2 Three-Way Interaction with `lm`

```
mod3 <- lm(y ~ x * z * w, data = dat)

jn3 <- JNK_lm(mod3,
```

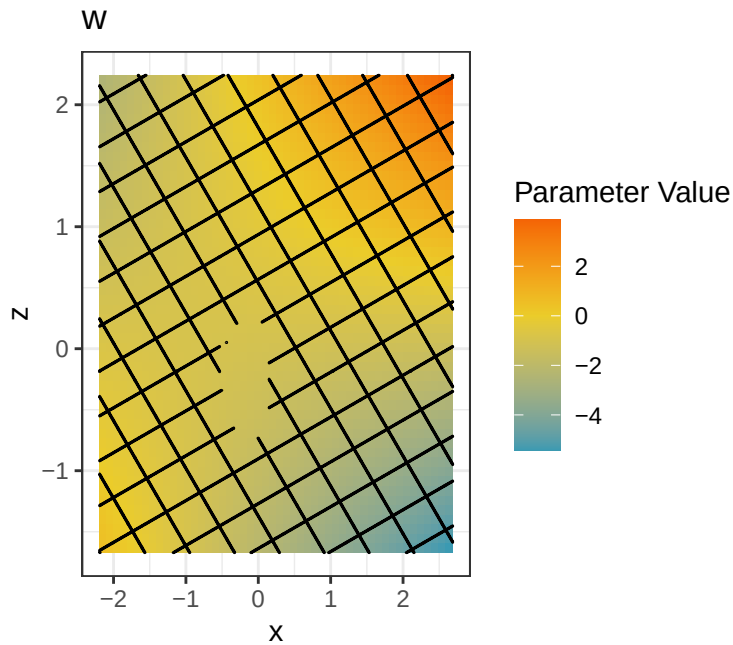
```
                theta_1    = "x",
                theta_2    = "z",
                theta_3    = "w",
                range_size = 50)
```



(a) Conditional effect of `x` across the (`z`, `w`) moderator grid.



(b) Conditional effect of `z` across the (`x`, `w`) moderator grid.



(c) Conditional effect of `w` across the (`x`, `z`) moderator grid.

Figure 2: Johnson–Neyman–Krause heatmaps for the three-way interaction among `x`, `z`, and `w` (`range_size = 50`, $\alpha = 0.05$). Each cell shows the conditional point estimate of the focal effect across the two-dimensional moderator grid; the diverging colour scale runs from blue (negative) through yellow (zero) to red (positive). Crosshatched cells indicate regions where the conditional effect is not statistically significant.

Setting `range_size = 50` produces a 50×50 grid for each of the three heatmaps (i.e., $3 \times 2{,}500$ conditional effect evaluations). When all three moderator ranges contain fewer than eight distinct values, the point estimate is printed in each cell; otherwise only the colour gradient communicates the magnitude. The ranges for each variable are by default extracted from the `lm`/`glm` object but can also be supplied directly, which is always necessary for `JNK_int()` and `JNK_siena()` (see the code chunk below).

4.3 Stochastic Actor-Oriented Models (`JNK_siena()`)

SAOMs estimated by **RSiena**'s `siena()` (formerly `siena07()`) are the primary application domain for which this package was developed, because interaction effects between network statistics and actor attributes are common yet difficult to interpret (T. A. B. Snijders et al. 2026). The interface mirrors `JNK_lm()` but uses integer indices referencing the position of each effect in the **RSiena** effects object[2].

```
# Two-way interaction
x2 <- JNK_siena(res,
                theta_1      = 6,
                theta_2      = 7,
                theta_int_12 = 8,
                theta_1vals  = -10:10,
                theta_2vals  = -10:10)

# Three-way extension
x3 <- JNK_siena(res,
                theta_1       = 6,
                theta_2       = 7,
                theta_3       = 5,
                theta_int_12  = 8,
                theta_int_13  = 10,
                theta_int_23  = 9,
                theta_int_123 = 11,
                theta_1vals   = -10:10,
                theta_2vals   = -10:10,
                theta_3vals   = 0:6,
                range_size    = 10)
```

The moderator value ranges (`theta_1vals`, `theta_2vals`, `theta_3vals`) should reflect the empirical support of the corresponding change statistics. By default `range_size = 500` equidis-

[2] While each effect in **RSiena** is named, within a given model, effects are not necessarily identifiable by their name but the same name might appear multiple times for different dependent variables or effect types

tant points are drawn from these ranges. The returned tables and plots are named as derived from the model using the 'name' column in the **RSiena** effects object.

4.4 Direct Use of `JNK_int()`

Because `JNK_int()` requires only a coefficient vector and a covariance matrix, it is straightforward to apply it to models not covered by the existing wrappers. For instance, for a mixed-effects model fitted with **lme4**:

```
library(lme4)
library(int3ract)

lme_mod <- lmer(y ~ x * z + (1 | group), data = dat)

all_coef <- fixef(lme_mod)
all_vcov <- as.matrix(vcov(lme_mod))
sel      <- c("x", "z", "x:z")

JNK_int(name  = c("x", "z"),
         coefs = all_coef[sel],
         covar = all_vcov[sel, sel],
         vals  = list(range(dat$x), range(dat$z)),
         range_size = 1000)
```

The same pattern applies to survival models (**survival**), ordinal regression (**MASS**), beta regression (**betareg**), and any other estimation that that provides parameter estimates and their covariances.

5 Bayesian Usage: `JNSienaBayes()`

For multi-group Bayesian SAOMs estimated with **multiSiena** (T. Snijders, Koskinen, and Ripley 2026; Koskinen and Snijders 2023), a series of conditional posterior distributions over the range of the moderator replaces the classic JN-framework. If the moderator range is given explicitly with fewer than eight unique values, distinct colors are used to plot each The function `JNSienaBayes()` automatically detects whether each required parameter has been estimated as a fixed effect (η; shared across groups) or as a random effect (μ; group-varying, with a hyper-parameter), and extracts the appropriate posterior draws.

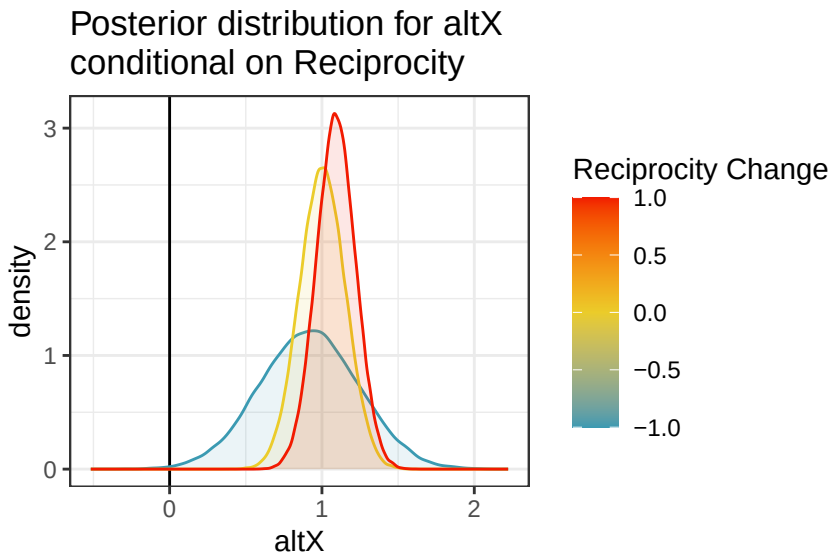
The returned object is a named list similar to those returned for frequentist interactions. Each table here contains the posterior mean, posterior SD, Bayesian p-value ($\Pr(\tilde{\theta}_X > 0 \mid$

data, model, prior)), and 2.5th and 97.5th percentiles of the conditional posterior for each moderator value.

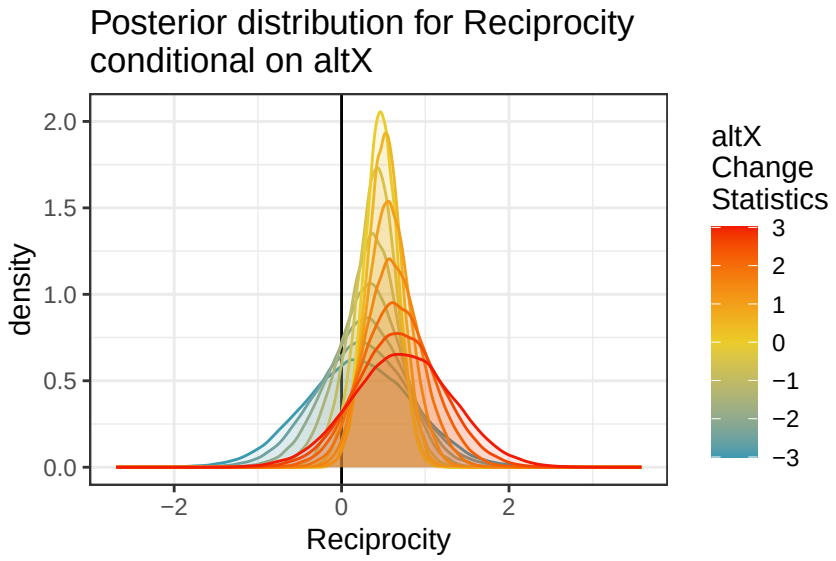
When `hyper_only = FALSE` and at least one of the involved parameters is randomly varying between groups, `JNSienaBayes()` additionally loops over all G groups and produces group-specific plots and tables, returned under the key `random_groups_effects`. This is useful for assessing heterogeneity of interaction effects across networks. Two additional arguments are helpful when many plots are made: `save = TRUE` will save each plot with `ggsave()`. To avoid cluttering your working directory, `folder` provides the location of a folder where the plots will be stored. If no input is provided, a new folder `int3ract JNplots` will be created (note that RStudio will ask you in the console whether such a folder is allowed to be created, if it does not exist yet).

For three-way interactions, the call is extended as usual. The `thresholds` argument (a vector of length 2, e.g., `c(0.05, 0.95)`) determines which cells of the posterior-mean heatmap are crosshatched: cells whose Bayesian p-value falls inside the interval $[\text{threshold}_1, \text{threshold}_2]$ are treated as non-significant.

```
jnb2 <- JNSienaBayes(sbo          = my_multiSiena_result,
                     theta_1      = 4,
                     theta_2      = 7,
                     theta_int_12 = 9,
                     theta_1_vals = c(-1,0,1),
                     theta_2_vals = seq(-3,3,0.5),
                     burnIn       = 500,
                     thin         = 2,
                     save         = FALSE)
```



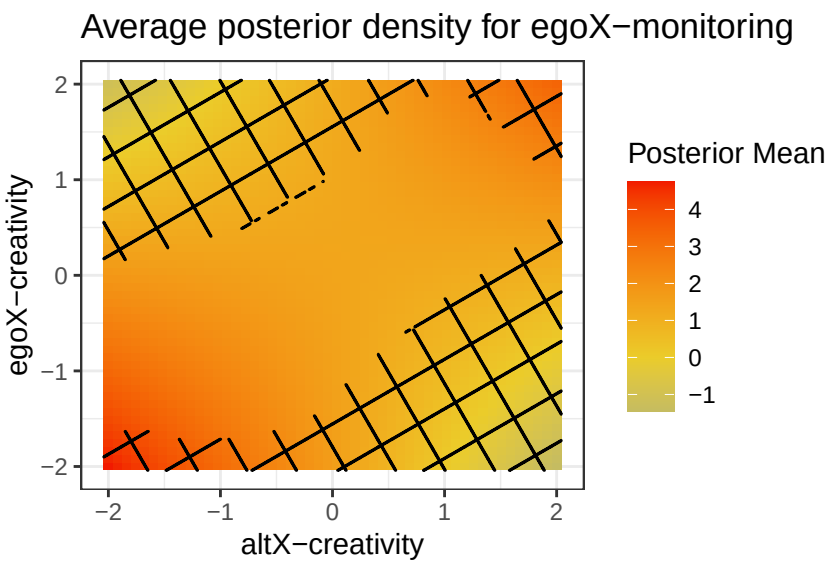
(a) Posterior densities for altX conditional on simultaneous changes in reciprocity. In this example, altX is always positive but has larger variance when reciprocity would decrease (i.e., a reciprocal tie to someone with negative values on the covariate would be broken, leading to -1 in recirpocity but overall an increase in altX).



(b) Posterior densities for reciprocity conditional on simultaneous changes in altX. Here, while generally positive, reciprocity only has a consistently positive effect on tie formation when altX changes moderately, around -1 to 1 (yellow curves). While reciprocity gets more positive with increasing values in the change of altX, the posterior variance increases as well.

Figure 3: Conditional posterior densities for the parameters for reciprocity and altX on a attribute.

```r
jnb3 <- JNSienaBayes(sbo           = my_multiSiena_result,
                     theta_1       = 4,
                     theta_2       = 10,
                     theta_3       = 9,
                     theta_int_12  = 21,
                     theta_int_13  = 22,
                     theta_int_23  = 11,
                     theta_int_123 = 23,
                     theta_1_vals  = seq(-2, 2, length.out = 50),
                     theta_2_vals  = seq(-2, 2, length.out = 50),
                     theta_3_vals  = seq(-2, 2, length.out = 50),
                     thresholds    = c(0.05, 0.95),
                     burnIn        = 500,
                     thin          = 2)
```



(a) Mean posterior densities for the effect of ego's self-monitoring on tie creation conditional on ego's and alter's creativity. In this example, self-monitoring has a positive effect on tie creation when both ego and alter have similar levels of creativity.

Figure 4: Figures for Bayesian 3-way interactions

6 Graphical Output and Customisation

All plots are standard **ggplot2** objects (Wickham 2016) and can be further modified using **ggplot2**'s layer system. The crosshatch pattern for non-significant regions is produced by

Table 2: Colour and pattern parameters available in all **int3ract** plotting functions. Default colour values are highlighted in their respective colour.

Parameter	Description	Default
sig_color	Ribbon fill for significant regions (2-way)	'seagreen3'
non_sig_color	Ribbon fill for non-significant regions (2-way)	'chocolate'
line_color	Point-estimate line colour (2-way)	'black'
color_low	Low end of diverging fill scale (heatmaps)	'#F05039'
color_mid	Midpoint colour of diverging fill scale	'white'
color_high	High end of diverging fill scale	'#000066'
color_grid	Crosshatch line colour	'black'
grid_density	Density of crosshatch lines	0.01
grid_spacing	Spacing between crosshatch lines	0.1
color_values	Within-cell text colour (heatmaps, 7 values)	'grey40'
crosshatch_non_sig	If TRUE, crosshatch marks the non-significant region	TRUE

ggpattern (Stewart 2022). Table 2 documents the colour and pattern parameters shared across all plotting functions. The colours for high, mid, and low point for continuous plots are taken from the `Zissou1` palette from the **wesanderson** package (Ram and Wickham 2023). Direct use of palettes is not supported but users can always overlay their own color palettes post-hoc by calling `scale_fill_gradient2()`/`scale_color_gradient2()` (for more and alternatives see `?ggplot2::scale_fill_gradient()`).

7 Summary and Outlook

int3ract provides a unified interface for the Johnson–Neyman technique and its three-way extension across a broad class of frequentist and Bayesian models. By exposing the core computation in a single model-agnostic function, `JNK_int()`, the package invites adoption beyond the two model classes for which dedicated wrappers currently exist. Future development plans include:

1. A wrapper for `merMod` objects from **lme4** to allow for randomly varying interactions.
2. A consideration for FDR-correction.
3. A general Bayesian application

Researchers are encouraged to report not only the JN boundaries but also the full plots as supplementary material, in keeping with calls for more transparent and complete reporting of interaction effects (Simonsohn, Simmons, and Nelson 2020).

8 Dependencies

To give due credit, next to the already listed packages, **int3ract** also depends on **scales**, **tidyr**, and **tibble** (Wickham, Pedersen, and Seidel 2025; Wickham, Vaughan, and Girlich 2025; Müller and Wickham 2026).

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